Drew University - CSCI 260 - Fall 2024
HW1 - Data Representation

HW Info.: Due on 09/09/2024 (M) by 11:59pm, 42 points
Submission: Submit as a single PDF file on Moodle.
Note: Please follow Drew University's Standard of Academic Integrity and refer to the course syllabus for late submission policy.

1. (16 points) Perform the following additions with 8-bit binary numbers. Then, compute the decimal values of the sums using the same corresponding representations (i.e., Signed Magnitude for part (a), and etc.). **Show all your work.**

   (a) Add the Signed Magnitude representation of 27 to the Signed Magnitude representation of -27.

   Sol. $(27)_{10} = (16+8+2+1)_{10} = (2^4+2^3+2^1+2^0)_{10} = (00011011)_2$; hence, in Signed Magnitude, $(-27)_{10} = (10011011)_2$.
   Thus, $(00011011)_2 + (10011011)_2 = (10110110)_2$.
   In Sign Magnitude, $(10110110)_2 = -(2^1+2^2+2^4+2^5)_{10} = -(2+4+16+32)_{10} = (-54)_{10}$.

   (b) Add the 1s' Complement representation of 34 to the 1s' Complement representation of -34.

   Sol. $(34)_{10} = (32+2)_{10} = (2^5+2^1)_{10} = (00100010)_2$; hence, in 1s' Complement, $(-34)_{10} = (11011101)_2$.
   Thus, $(00100010)_2 + (11011101)_2 = (11111111)_2$.
   In 1s' Complement, $(11111111)_2 = -(00000000)_2 = (-0)_{10}$.

   (c) Add the 2's Complement representation of 59 to the 2's Complement representation of 59.

   Sol. $(59)_{10} = (32+16+8+2+1)_{10} = (2^5+2^4+2^3+2^1+2^0)_{10} = (00111011)_2$.
   Thus, $(00111011)_2 + (00111011)_2 = 2 * (00111011)_2 = (01110110)_2$ as a left-shift where the m.s.b exceeds the length and was ignored. In 2's Complement, $(01110110)_2 = (2^1+2^2+2^4+2^5+2^6)_{10} = (2+4+16+32+64)_{10} = (118)_{10}$.

   (d) Add Unsigned binary representation of 153 to the Unsigned binary representation of 102.
   Sol. $(153)_{10} = (128+16+8+1)_{10} = (2^7+2^4+2^3+2^0)_{10} = (10011001)_2$; and
   $(102)_{10} = (64+32+4+2)_{10} = (2^6+2^5+2^2+2^1)_{10} = (01100110)_2$.
   Thus, $(10011001)_2 + (01100110)_2 = (11111111)_2$.
   In decimal, $(11111111)_2 = (2^0+2^1+2^2+2^3+2^4+2^5+2^6+2^7)_{10} = (255)_{10}$. OR, since this is the largest 8-bit unsigned binary number we can obtain, this is $(2^8-1)_{10} = (256-1)_{10} = (255)_{10}$.

2. (10 points) Compute the following logical expressions. **Show all your work.**

   (a) NOT(1011) OR NOT(1100) = 0100 OR 0011 = 0111

   (b) NOT(1000 AND (1100 OR 0101)) = NOT(1000 AND 1101) = NOT(1000) = 0111

   (c) NOT(NOT(1101)) = NOT(0010) = 1101

   (d) (0110 OR 0000) AND 1111 = 0110 AND 1111 = 0110

   (e) NOT(xDEFA) AND NOT(xFFFF) in hexadecimal = NOT(1101 1110 1111 1010) AND NOT(1111 1111 1111 1111) = 0010 0001 0000 0101 AND 0000 0000 0000 0000 = 0000 0000 0000 0000 = x0000

3. (16 points) Consider four hexadecimal numbers: 0x434F, 0x4D50, 0x5554, and 0x4552. What values do they represent if treated as each of the four data types shown? For the first 3 rows, you should provide decimal values; and for the last row, you should provide text values. **Show all your work.**

	0x434F	0x4D50	x5554	0x4552
Unsigned Binary	$(17231)_{10}$	$(19792)_{10}$	$(21844)_{10}$	$(17746)_{10}$
1's Complement	$(17231)_{10}$	$(19792)_{10}$	$(21844)_{10}$	$(17746)_{10}$
2's Complement	$(17231)_{10}$	$(19792)_{10}$	$(21844)_{10}$	$(17746)_{10}$
ASCII String	"CO"	"MP"	"UT"	"ER"

Sol. 0x434F reads "CO" in ASCII. It is $(0100\ 0011\ 0100\ 1111)_2$ in binary. Note that this is a positive number in any of the 3 representations. Hence, we have $(2^0 + 2^1 + 2^2 + 2^3 + 2^6 + 2^8 + 2^9 + 2^{14})_{10} = (17231)_{10}$.

0x4D50 reads "MP" in ASCII. It is $(0100\ 1101\ 0101\ 0000)_2$ in binary. Again, a positive number in any of the 3 representations. Hence, we have $(2^4 + 2^6 + 2^8 + 2^{10} + 2^{11} + 2^{14})_{10} = (19792)_{10}$

0x5554 reads "UT" in ASCII. It is $(0101\ 0101\ 0101\ 0100)_2$ in binary. Again, positive. Hence, we have $(2^2 + 2^4 + 2^6 + 2^8 + 2^{10} + 2^{12} + 2^{14})_{10} = (21844)_{10}$.

0x4552 reads "ER" in ASCII. It is $(0100\ 0101\ 0101\ 0010)_2$ in binary. Again, positive. Hence, we have $(2^1 + 2^4 + 2^6 + 2^8 + 2^{10} + 2^{14})_{10} = (17746)_{10}$.